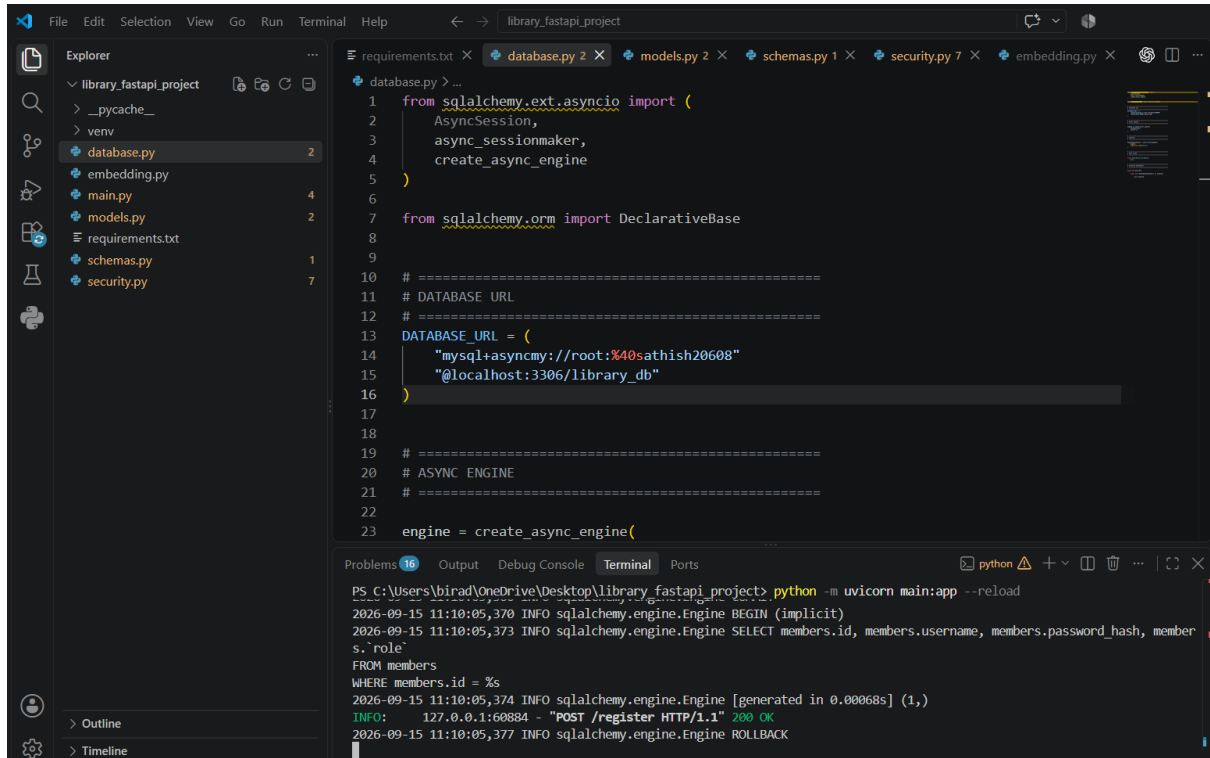


Sathish kumar _2520090008

Practical – 5 Library project



```
File Edit Selection View Go Run Terminal Help
library_fastapi_project

Explorer
library_fastapi_project
  > __pycache__
  > venv
  database.py 2
  embedding.py
  main.py 4
  models.py 2
  requirements.txt
  schemas.py 1
  security.py 7

database.py
1 from sqlalchemy.ext.asyncio import (
2     AsyncSession,
3     async_sessionmaker,
4     create_async_engine
5 )
6
7 from sqlalchemy.orm import DeclarativeBase
8
9
10 # =====
11 # DATABASE URL
12 # =====
13 DATABASE_URL = (
14     "mysql+asyncmy://root:%40sathish20608"
15     "@localhost:3306/library_db"
16 )
17
18
19 # =====
20 # ASYNC ENGINE
21 # =====
22
23 engine = create_async_engine(

Problems 16 Output Debug Console Terminal Ports
python
PS C:\Users\birad\OneDrive\Desktop\library_fastapi_project> python -m uvicorn main:app --reload
2026-09-15 11:10:05,370 INFO sqlalchemy.engine.Engine BEGIN (implicit)
2026-09-15 11:10:05,373 INFO sqlalchemy.engine.Engine SELECT members.id, members.username, members.password_hash, member
s.`role`
FROM members
WHERE members.id = %s
2026-09-15 11:10:05,374 INFO sqlalchemy.engine.Engine [generated in 0.00068s] (1,)
INFO: 127.0.0.1:60884 - "POST /register HTTP/1.1" 200 OK
2026-09-15 11:10:05,377 INFO sqlalchemy.engine.Engine ROLLBACK
```

My sql

MySQL Workbench

MySQL Model x library project x

File Edit View Query Database Server Tools Scripting Help

Navigator

Query 1 x

Limit to 1000 rows

17
18 • USE library_db;
19
20
21
22 -- 3. DROP OLD TABLES (OPTIONAL CLEAN START)
23 --
24
25 • DROP TABLE IF EXISTS books;

Result Grid

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI		auto_increment
username	varchar(100)	NO	UNI		
password_hash	varchar(255)	NO			
role	varchar(30)	NO		member	

Administration Schemas

Information

No object selected

Result 7 Result 8 x Result 9 members 10 books 11

Output

Action Output

#	Time	Action	Message
21	10:58:35	CREATE TABLE books (id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(250) NOT NULL, ...	0 row(s) affected
22	10:58:35	SHOW TABLES	2 row(s) returned
23	10:58:36	DESCRIBE members	4 row(s) returned
24	10:58:36	DESCRIBE books	8 row(s) returned
25	10:58:36	SELECT * FROM members LIMIT 0, 1000	0 row(s) returned
26	10:58:36	SELECT * FROM books LIMIT 0, 1000	0 row(s) returned

Object Info Session

Post:

GET / Home

POST /register Register

Parameters

No parameters

Request body **required**

application/json

Edit Value | Schema

```
{
  "username": "lct",
  "password": "lct@143"
}
```

Execute Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/register' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "username": "lst",
    "password": "lst@143"
  }'
```

Request URL

http://127.0.0.1:8000/register

Server response

Code Details

200

Response body

```
{
  "message": "Member registered successfully",
  "username": "lst",
  "role": "member"
}
```

Response headers

```
content-length: 77
content-type: application/json
date: Tue, 15 Sep 2026 05:51:35 GMT
server: uvicorn
```

Responses

Code Description Links

In SQL user added :

MySQL Workbench

library project x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

- 2520090008_sathish
- 2520090008_sathishkumar
- bookflow_db
- collegedb
- klu
- library_db**
 - Tables
 - Views
 - Stored Procedures
 - Functions
- sathish_2520090008
- sys
- university

Administration Schemas

Information

Schema: library_db

Query 1

```
1  --
2  -- LIBRARY FASTAPI PROJECT
3  -- FINAL MYSQL DATABASE SETUP
4  --
5
6
7  --
8  -- 1. CREATE DATABASE
9
```

Result Grid

	id	username	password_hash	role
1	lst	\$2b\$12\$JXt6...	me...	

Result 1 Result 2 Result 3 members 4 x books 5

Output

Action Output

#	Time	Action	Message
6	11:20:35	CREATE TABLE books (id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(250) NOT NULL, ...	0 row(s) affected
7	11:20:52	SHOW TABLES	2 row(s) returned
8	11:20:52	DESCRIBE members	4 row(s) returned
9	11:20:52	DESCRIBE books	8 row(s) returned
10	11:20:52	SELECT * FROM members LIMIT 0, 1000	0 row(s) returned
11	11:20:52	SELECT * FROM books LIMIT 0, 1000	0 row(s) returned

Object Info Session

POST

/books

Create Book

Parameters

Cancel

No parameters

Request body required

application/json

Edit Value

Schema

```
{
  "title": "string",
  "author": "string",
  "isbn": "stringstrings",
  "price": 1,
  "published date": "2026-09-15",
  "description": "string"
}
```

Execute

Clear

Responses

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/books' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJsc3QlLCJyb2x1IjoibWVYVWVYIiwiaWF0Ijoi2026-09-15T00:00:00Zy6jeBk5Px111518u8obV6xPLIz8oAqydbuBUPh4kQ' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "string",
    "author": "string",
    "isbn": "stringstrings",
    "price": 1,
    "published date": "2026-09-15",
    "description": "string"
  }'
```

Request URL

http://127.0.0.1:8000/books

Server response

Code

Details

201

Response body

```
{
  "message": "Book added successfully",
  "book id": 1,
  "added by": "1st"
}
```

Download

Response headers

```
content-length: 66
content-type: application/json
date: Tue, 15 Sep 2026 06:05:27 GMT
server: uvicorn
```

Responses

In sql:



Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/books/search?q=python&limit=5' \
-H 'accept: application/json' \
-H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIjOiJsc3QlICJyY2kiOiJoIiwiaWF0IjoxNjc5NDUxOTgwQWQwPyG6eBbS:Px1Z1S18NaVbnV6PzI78aQndqbUBUMHkQ'
```

Request URL

`http://127.0.0.1:8000/books/search?q=python&limit=5`

Server response

Code	Details
------	---------

200

Response body

```
{
  "query": "python",
  "results": [
    {
      "id": 1,
      "title": "string",
      "author": "string",
      "isbn": "stringstrings",
      "price": 1,
      "published date": "2026-09-15",
      "description": "string",
      "similarity": 0.015782309219735828
    }
  ]
}
```

Response headers

```
content-length: 196
content-type: application/json
date: Tue, 15 Sep 2026 06:08:04 GMT
server: uvicorn
```

```
content-length: 196
content-type: application/json
date: Tue, 15 Sep 2026 06:08:04 GMT
server: uvicorn
```

Responses

Code	Description
------	-------------

Links

200

Successful Response

No links

Media type

application/json

Controls Accept header.

Example Value | Schema

"string"

422

Validation Error

No links

Media type

```
application/json
```

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```